

# Random variables

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## Random variables

**Definition 0.1** (Random variable). A **random variable** is a [function](#) from a [sample space](#)  $\Omega$  to the real numbers  $\mathbb{R}$ : it assigns a number to each possible outcome of a random experiment.

*Remark.* We use uppercase letters ( $X, Y, Z, \dots$ ) for random variables, and lowercase letters ( $x, y, z, \dots$ ) for their realized (observed) values. We write  $\{X = x\}$  for the event  $\{\omega \in \Omega : X(\omega) = x\}$ , and similarly for  $\{X \leq x\}$  and other conditions on  $X$ . A fully rigorous definition also requires the function to be *measurable*, so that every such set is an event with a probability; that requirement never binds in these notes.

See also: [https://en.wikipedia.org/wiki/Random\\_variable](https://en.wikipedia.org/wiki/Random_variable)

**Example 0.1** (Number of heads in two coin flips). Flip a fair coin twice. The sample space is  $\Omega = \{HH, HT, TH, TT\}$ , each outcome with probability  $1/4$ . Let  $X$  be the number of heads:

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0$$

Then  $\{X = 1\} = \{HT, TH\}$ , so  $\Pr(X = 1) = 1/4 + 1/4 = 1/2$ .

**Definition 0.2** (Range of a random variable). The **range** of a random variable  $X$ , denoted  $\mathcal{R}(X)$ , is the set of values  $X$  can take:

$$\mathcal{R}(X) \stackrel{\text{def}}{=} \{X(\omega) : \omega \in \Omega\}$$

*Remark.* The range of  $X$  is the **image** of  $X$  as a function on  $\Omega$ . For a general function, these notes say “image”, because some sources use “range” for the codomain instead; for a random variable, “range” is the standard term.

**Example 0.2** (Range of the number of heads). In Example 0.1,  $\mathcal{R}(X) = \{0, 1, 2\}$ .

**Definition 0.3** (Discrete random variable). A random variable  $X$  is **discrete** if its **range**  $\mathcal{R}(X)$  is **countable**.

**Example 0.3** (Discrete random variables). The number of heads in Example 0.1 is discrete, with a finite range. A count with no upper bound, such as the number of emergency-department visits in a day, is discrete with a countably infinite range,  $\{0, 1, 2, \dots\}$ .

**Definition 0.4** (Continuous random variable). A random variable  $X$  is **continuous** if  $\Pr(X = x) = 0$  for every  $x \in \mathbb{R}$ .

**Definition 0.5** (Uniform distribution). A random variable  $X$  has the **uniform distribution** on an interval  $[\alpha, \beta]$ , with  $\alpha < \beta$ , written  $X \sim \text{Uniform}(\alpha, \beta)$ , if the probability that  $X$  falls in any subinterval of  $[\alpha, \beta]$  is proportional to the length of that subinterval:

$$\Pr(a \leq X \leq b) = \frac{b - a}{\beta - \alpha} \quad \text{for all } \alpha \leq a \leq b \leq \beta$$

**Example 0.4** (A uniformly distributed random variable). Let  $X \sim \text{Uniform}(0, 1)$  (Definition 0.5), so  $\Pr(a \leq X \leq b) = b - a$  for all  $0 \leq a \leq b \leq 1$ . Taking  $a = b = x$  gives  $\Pr(X = x) = x - x = 0$  for every  $x \in [0, 1]$ . Taking  $a = 0$  and  $b = 1$  gives  $\Pr(0 \leq X \leq 1) = 1$ , so by the **complement rule**  $X$  falls outside  $[0, 1]$  with probability 0, and  $\Pr(X = x) = 0$  for every  $x$  outside  $[0, 1]$  as well. So  $X$  is continuous. Its range,  $[0, 1]$ , is uncountable.

**Example 0.5** (A random variable that is neither discrete nor continuous). Some random variables are neither discrete nor continuous: a time to event that equals exactly 0 with positive probability, and otherwise spreads over  $(0, \infty)$ , is one. For instance, let  $U \sim \text{Uniform}(0, 1)$  (Definition 0.5), and let  $T = 1/(1 - U) - 2$  when  $1/2 < U < 1$ , and  $T = 0$  otherwise. On  $\{1/2 < U < 1\}$ ,  $T$  is increasing in  $U$  and takes values in  $(0, \infty)$ , and solving  $t = 1/(1 - u) - 2$  for  $u$  gives  $u = 1 - 1/(t + 2)$ . So  $\{T > 0\} = \{1/2 < U < 1\}$ , and  $\{T = t\} = \{U = 1 - 1/(t + 2)\}$  for each  $t > 0$ . Since  $\Pr(U = 1/2) = \Pr(U = 1) = 0$

(Example 0.4):

$$\begin{aligned}
\Pr(T > 0) &= \Pr(1/2 < U < 1) && (\{T > 0\} = \{1/2 < U < 1\}) \\
&= \Pr(1/2 \leq U \leq 1) - \Pr(U = 1/2) - \Pr(U = 1) && (\text{additivity}) \\
&= \frac{1}{2} - 0 - 0 && (\text{definition of the uniform distribution}) \\
&= \frac{1}{2} && (\text{simplify})
\end{aligned}$$

$T$  is not continuous, because  $\Pr(T = 0) = 1 - \Pr(T > 0) = 1/2$  by the [complement rule](#).  $T$  is not discrete either. If it were, its range would be countable, so  $\{T > 0\}$  would be the disjoint union of the countably many events  $\{T = t\}$  with  $t \in \mathcal{R}(T)$  and  $t > 0$ , and:

$$\begin{aligned}
\Pr(T > 0) &= \sum_{t \in \mathcal{R}(T), t > 0} \Pr(T = t) && (\text{countable additivity}) \\
&= \sum_{t \in \mathcal{R}(T), t > 0} \Pr(U = 1 - \frac{1}{t+2}) && (\{T = t\} = \{U = 1 - 1/(t+2)\}) \\
&= 0 && (U \text{ is continuous})
\end{aligned}$$

which contradicts  $\Pr(T > 0) = 1/2$ .

## Characteristics of probability distributions

**Definition 0.6** (Probability mass function (PMF)). If  $X$  is a [discrete random variable](#), the **probability mass function** of  $X$  at value  $x$ , denoted  $f(x)$ ,  $f_X(x)$ ,  $P(x)$ ,  $P_X(x)$ , or  $P(X = x)$ , is the probability that  $X$  takes exactly the value  $x$ :

$$P(X = x) \stackrel{\text{def}}{=} \Pr(\{X = x\})$$

*Remark.* See also [https://en.wikipedia.org/wiki/Probability\\_mass\\_function](https://en.wikipedia.org/wiki/Probability_mass_function)

**Example 0.6** (PMF of the number of heads). For the number of heads  $X$  in two fair coin flips (Example 0.1):

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{4}$$

**Definition 0.7** (Bernoulli distribution). A [discrete](#) random variable  $X$  has the **Bernoulli distribution** with parameter  $\pi \in [0, 1]$ , written  $X \sim \text{Ber}(\pi)$ , if:

$$\begin{aligned}\Pr(X = x) &\stackrel{\text{def}}{=} 1_{x \in \{0,1\}} \pi^x (1 - \pi)^{1-x} \\ &= \begin{cases} \pi, & x = 1 \\ 1 - \pi, & x = 0 \end{cases}\end{aligned}$$

**Example 0.7** (The first of two coin flips). In Example 0.1, let  $X_1$  indicate heads on the first flip, so  $X_1(HH) = X_1(HT) = 1$  and  $X_1(TH) = X_1(TT) = 0$ . Then  $P(X_1 = 1) = \Pr(\{HH, HT\}) = 1/2$  and  $P(X_1 = 0) = \Pr(\{TH, TT\}) = 1/2$ , so  $X_1 \sim \text{Ber}(1/2)$ .

**Definition 0.8** (Probability density function (PDF)). If  $X$  is a **continuous random variable**, a **probability density function** of  $X$ , denoted  $f(x)$ ,  $f_X(x)$ ,  $p(x)$ ,  $p_X(x)$ , or  $p(X = x)$ , is a function  $f$  that satisfies:

- $f(x) \geq 0$  for every  $x$ .
- The integral of  $f$  over any interval is the **probability** that  $X$  falls in that interval:

$$\Pr(a \leq X \leq b) = \int_a^b f(x) dx \quad \text{for all } a \leq b$$

*Remark.* See also [https://en.wikipedia.org/wiki/Probability\\_density\\_function#Formal\\_definition](https://en.wikipedia.org/wiki/Probability_density_function#Formal_definition)

**Example 0.8** (Density of a uniform random variable). Let  $X \sim \text{Uniform}(0, 1)$  (Example 0.4), and let  $f(x) = 1$  for  $x \in [0, 1]$  and  $f(x) = 0$  otherwise. For  $0 \leq a \leq b \leq 1$ :

$$\begin{aligned}\int_a^b f(x) dx &= \int_a^b 1 dx && (f = 1 \text{ on } [0, 1]) \\ &= b - a && (\text{integrate}) \\ &= \Pr(a \leq X \leq b) && (\text{definition of the uniform distribution})\end{aligned}$$

An interval reaching outside  $[0, 1]$  adds nothing to either side:  $f = 0$  there, and  $X$  falls outside  $[0, 1]$  with probability 0 (Example 0.4). So  $f$  is a density of  $X$ .

**Theorem 0.1** (A density is not unique). *If  $f$  is a **density** of a continuous random variable  $X$ , and  $g \geq 0$  differs from  $f$  at only finitely many points, then  $g$  is also a density of  $X$ .*

**i** Proof

*Proof.* For  $a \leq b$ , the difference  $g - f$  is 0 at all but finitely many points, so:

$$\begin{aligned}\int_a^b g(x) dx &= \int_a^b f(x) dx + \int_a^b (g(x) - f(x)) dx && \text{(linearity of the integral)} \\ &= \int_a^b f(x) dx + 0 && \text{(a function that is 0 at all but finitely many points integrates to 0)} \\ &= \Pr(a \leq X \leq b) && (f \text{ is a density of } X)\end{aligned}$$

Together with  $g \geq 0$ , this is Definition 0.8 for  $g$ . □

*Remark.* So the value of a density at a single point carries no probability by itself. These notes use the version that is continuous wherever possible.

**Theorem 0.2** (Some continuous random variables have no density). *There exist continuous random variables that have no [probability density function](#).*

*Remark.* The proof needs measure theory beyond these notes. The standard example is the [Cantor distribution](#), which puts no probability on any single point, but puts all of its probability on a set of total length 0, so any candidate density would integrate to 0 over a set of probability 1.

Every continuous random variable in these notes also has a [probability density function](#). Continuous random variables without a density do not arise in applications.

**Definition 0.9** (Normal distribution). A random variable  $X$  has the **normal distribution** (or **Gaussian distribution**) with mean parameter  $\mu \in \mathbb{R}$  and variance parameter  $\sigma^2 > 0$ , written  $X \sim N(\mu, \sigma^2)$ , if  $X$  is continuous with [density](#):

$$p(X = x) \stackrel{\text{def}}{=} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in \mathbb{R}$$

**Example 0.9** (Normal densities at their centers). For  $X \sim N(0, 1)$ , the density at  $x = 0$  is:

$$\begin{aligned}
p(X = 0) &= \frac{1}{1 \cdot \sqrt{2\pi}} e^{-\frac{(0-0)^2}{2 \cdot 1}} && \text{(normal density with } \mu = 0, \sigma^2 = 1) \\
&= \frac{1}{\sqrt{2\pi}} && (e^0 = 1) \\
&\approx 0.399 && \text{(evaluate)}
\end{aligned}$$

For  $X \sim N(0, 0.1^2)$ , the same steps give  $p(X = 0) = 1/(0.1\sqrt{2\pi}) \approx 3.99$ , a density greater than 1.

*Remark.* A density is not a probability: it can exceed 1, as the second density in Example 0.9 does.

**Theorem 0.3** (The normal density integrates to 1, with mean  $\mu$  and variance  $\sigma^2$ ). *If  $X \sim N(\mu, \sigma^2)$  (Definition 0.9), then:*

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = 1$$

and  $X$  has *mean*  $E[X] = \mu$  and *variance*  $\text{Var}(X) = \sigma^2$ .

*Remark.* The proof evaluates three integrals by calculus, one of them the Gaussian integral; it is omitted here (Casella and Berger 2002). These facts are where the parameters' names come from.

**Definition 0.10** (Standard normal distribution). The **standard normal distribution** is the *normal distribution* with mean parameter 0 and variance parameter 1,  $N(0, 1)$ .

**Example 0.10** (Half of a standard normal lies below 0). If  $Z$  has the standard normal distribution, its density  $p(Z = z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$  takes the same value at  $z$  and  $-z$ , so  $\Pr(Z \leq 0) = \Pr(Z \geq 0)$ . Those two probabilities sum to 1, because  $\Pr(Z = 0) = 0$  for a continuous  $Z$ , so each is  $\frac{1}{2}$ .

**Definition 0.11** (Exponential distribution). A random variable  $T$  has the **exponential distribution** with rate  $\lambda > 0$  if  $T$  is continuous with *density*:

$$f(t) \stackrel{\text{def}}{=} \begin{cases} \lambda e^{-\lambda t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

**Example 0.11** (Probability of an event within one time unit). Let  $T$  be exponential with rate  $\lambda$ . The probability that  $T$  falls in  $[0, 1]$  is:

$$\begin{aligned}\Pr(0 \leq T \leq 1) &= \int_0^1 \lambda e^{-\lambda t} dt \quad (\text{definition of a density}) \\ &= [-e^{-\lambda t}]_{t=0}^1 \quad (\text{antiderivative of } \lambda e^{-\lambda t}) \\ &= 1 - e^{-\lambda} \quad (\text{evaluate at the bounds})\end{aligned}$$

With  $\lambda = 0.5$ , for example,  $\Pr(0 \leq T \leq 1) = 1 - e^{-0.5} \approx 0.393$ .

**Definition 0.12** (Cumulative distribution function (CDF)). For a random variable  $X$  (discrete or continuous), its **cumulative distribution function** is:

$$F(t) \stackrel{\text{def}}{=} \Pr(X \leq t), \quad t \in \mathbb{R}.$$

*Remark.* The CDF is always defined, whether  $X$  is discrete or continuous. See also [https://en.wikipedia.org/wiki/Cumulative\\_distribution\\_function](https://en.wikipedia.org/wiki/Cumulative_distribution_function)

**Example 0.12** (CDF of the number of heads). Summing the PMF in Example 0.6 over values at or below  $t$ :

$$F(t) = \begin{cases} 0, & t < 0 \\ \frac{1}{4}, & 0 \leq t < 1 \\ \frac{3}{4}, & 1 \leq t < 2 \\ 1, & t \geq 2 \end{cases}$$

**Theorem 0.4** (Properties of a CDF). The *CDF*  $F$  of any random variable  $X$  is non-decreasing, and:

$$\lim_{t \rightarrow -\infty} F(t) = 0, \quad \lim_{t \rightarrow \infty} F(t) = 1$$

### **i** Proof

*Proof. Non-decreasing.* For  $s < t$ ,  $\{X \leq t\}$  is the disjoint union of  $\{X \leq s\}$  and  $\{s < X \leq t\}$ , so *finite additivity* gives:

$$\begin{aligned}
F(t) &= \Pr(X \leq s) + \Pr(s < X \leq t) && \text{(additivity, and the definition of the CDF)} \\
&\geq \Pr(X \leq s) && \text{(probabilities are non-negative)} \\
&= F(s) && \text{(definition of the CDF)}
\end{aligned}$$

*Limit at  $-\infty$ .* The events  $A_n = \{X \leq -n\}$ ,  $n = 1, 2, \dots$ , are decreasing, and their intersection is empty, because every outcome  $\omega$  has  $X(\omega) > -n$  once  $n > -X(\omega)$ . By [continuity of probability](#),  $F(-n) = \Pr(A_n) \rightarrow 0$ . For  $t \leq -n$ ,  $0 \leq F(t) \leq F(-n)$ , because  $F$  is non-decreasing, so  $F(t) \rightarrow 0$  as  $t \rightarrow -\infty$ .

*Limit at  $\infty$ .* The events  $B_n = \{X > n\}$  are decreasing, and their intersection is empty, because every outcome  $\omega$  has  $X(\omega) \leq n$  once  $n \geq X(\omega)$ . By continuity of probability,  $\Pr(B_n) \rightarrow 0$ , so, by the [complement rule](#):

$$\begin{aligned}
F(n) &= 1 - \Pr(X > n) && \text{(complement rule)} \\
&\rightarrow 1 - 0 && (\Pr(B_n) \rightarrow 0)
\end{aligned}$$

For  $t \geq n$ ,  $F(n) \leq F(t) \leq 1$ , because  $F$  is non-decreasing, so  $F(t) \rightarrow 1$  as  $t \rightarrow \infty$ .  $\square$

**Theorem 0.5** (The CDF determines the distribution). *If random variables  $X$  and  $Y$  have the same [CDF](#), then  $\Pr(X \in B) = \Pr(Y \in B)$  for every [Borel set](#)  $B \subseteq \mathbb{R}$ , which includes every interval, and every set built from intervals by countably many unions, intersections, and complements.*

*Remark.* The proof needs measure theory beyond these notes (Billingsley 1995); see also [https://en.wikipedia.org/wiki/Cumulative\\_distribution\\_function](https://en.wikipedia.org/wiki/Cumulative_distribution_function).

**Definition 0.13** (Quantile function (population inverse CDF)). For a random variable  $X$  with [cumulative distribution function \(CDF\)](#)  $F$ , its population **quantile function** (generalized inverse of  $F$ ) is:

$$Q(p) \stackrel{\text{def}}{=} \inf\{t : F(t) \geq p\}, \quad 0 < p < 1.$$

*Remark.* The restriction to  $0 < p < 1$  keeps  $Q(p)$  finite: for  $p = 1$ , the set  $\{t : F(t) \geq 1\}$  is empty whenever  $F(t) < 1$  for every  $t$ , as for a [normal distribution](#), and the infimum of the empty set is  $+\infty$ .

**Example 0.13** (Median of the number of heads). From Example [0.12](#),  $F(t) \geq 0.5$  first holds at  $t = 1$ , so the median of the number of heads is  $Q(0.5) = 1$ .

**Theorem 0.6** (A density integrates to the CDF, and is its derivative). *If  $X$  is a continuous random variable with density  $f$  and CDF  $F$ , then for every  $t$ :*

$$F(t) = \int_{-\infty}^t f(x) dx$$

*and at every  $t$  where  $f$  is continuous:*

$$f(t) = \frac{\partial}{\partial t} F(t)$$

### **i** Proof

*Proof.* For any  $a < t$ ,  $\{X \leq t\}$  is the disjoint union of  $\{X \leq a\}$  and  $\{a < X \leq t\}$ , and  $\Pr(X = a) = 0$  because  $X$  is continuous. So:

$$\begin{aligned} F(t) &= \Pr(X \leq a) + \Pr(a < X \leq t) && \text{(additivity, and the definition of the CDF)} \\ &= F(a) + \Pr(a \leq X \leq t) && \text{(definition of the CDF; } \Pr(X = a) = 0) \\ &= F(a) + \int_a^t f(x) dx && \text{(definition of the density)} \end{aligned}$$

Letting  $a \rightarrow -\infty$ ,  $F(a) \rightarrow 0$  (Theorem 0.4), which gives  $F(t) = \int_{-\infty}^t f(x) dx$ . The fundamental theorem of calculus then gives  $\frac{\partial}{\partial t} \int_{-\infty}^t f(x) dx = f(t)$  at every  $t$  where  $f$  is continuous.  $\square$

**Corollary 0.1** (A CDF has its density as derivative at all but finitely many points). *If  $X$  is a continuous random variable with CDF  $F$  and a density  $f$  that is continuous at all but finitely many points, then  $F$  is differentiable with  $F' = f$  at all but finitely many points.*

### **i** Proof

*Proof.* At every point  $t$  where  $f$  is continuous, Theorem 0.6 gives  $F'(t) = f(t)$ , and by assumption those are all but finitely many points.  $\square$

*Remark.* Every density in these notes is continuous at all but finitely many points, so Corollary 0.1 applies to each of them.

**Example 0.14** (CDF and density of a uniform random variable). For  $X$  uniform on  $[0, 1]$  with the density  $f$  of Example 0.8, and  $t \in [0, 1]$ :

$$\begin{aligned}
F(t) &= \int_{-\infty}^t f(x) dx && \text{(the CDF is the integral of the density)} \\
&= \int_{-\infty}^0 0 dx + \int_0^t 1 dx && \text{(split at 0, and substitute } f) \\
&= t && \text{(integrate)}
\end{aligned}$$

For  $t \in (0, 1)$ , where  $f$  is continuous,  $\frac{\partial}{\partial t} F(t) = 1 = f(t)$ . At  $t = 1$ ,  $f$  jumps from 1 to 0, and  $F$  has no derivative: its slope is 1 on the left and 0 on the right.

**Theorem 0.7** (A piecewise-smooth CDF has its derivative as a density). *If the CDF  $F$  of a random variable  $X$  is continuous everywhere, and has a continuous derivative at all but finitely many points, then  $X$  is continuous, and  $F'$  (given any values at the finitely many exceptional points) is a density of  $X$ .*

### **i** Proof

*Proof.* For events  $A \subseteq B$ ,  $B$  is the disjoint union of  $A$  and  $B \setminus A$ , so additivity gives  $\Pr(A) \leq \Pr(B)$ . For any  $x$  and  $\Delta > 0$ ,  $\{X = x\} \subseteq \{x - \Delta < X \leq x\}$ , so:

$$\begin{aligned}
0 &\leq \Pr(X = x) && \text{(probabilities are non-negative)} \\
&\leq \Pr(x - \Delta < X \leq x) && (\{X = x\} \subseteq \{x - \Delta < X \leq x\}) \\
&= F(x) - F(x - \Delta) && \text{(additivity, and the definition of the CDF)}
\end{aligned}$$

and  $F(x) - F(x - \Delta) \rightarrow 0$  as  $\Delta \downarrow 0$ , because  $F$  is continuous. So  $\Pr(X = x) = 0$  for every  $x$ , and  $X$  is continuous. Then, for  $a \leq b$ :

$$\begin{aligned}
\Pr(a \leq X \leq b) &= \Pr(a < X \leq b) \quad (\Pr(X = a) = 0) \\
&= F(b) - F(a) && \text{(additivity, and the definition of the CDF)} \\
&= \int_a^b F'(x) dx && \text{(fundamental theorem of calculus)}
\end{aligned}$$

The last step applies the fundamental theorem of calculus on each piece of  $[a, b]$  between exceptional points, where  $F'$  is continuous, and joins the pieces using the continuity of  $F$  at the exceptional points. Finally,  $F' \geq 0$  wherever it exists, because  $F$  is non-decreasing (Theorem 0.4).  $\square$

**Example 0.15** (A density from a CDF with a corner). Let  $X$  have CDF  $F(t) = 0$  for  $t < 0$ ,  $F(t) = t^2$  for  $0 \leq t \leq 1$ , and  $F(t) = 1$  for  $t > 1$ . This  $F$  is continuous everywhere, and has a continuous derivative everywhere except  $t = 1$ , where its slope drops from 2 to 0. By Theorem 0.7,  $X$  is continuous with density  $f(t) = 2t$  for  $0 \leq t < 1$  and  $f(t) = 0$

otherwise. As a check,  $\int_0^1 2t \, dt = 1$ .

**Theorem 0.8** (A density as a limit of interval probabilities). *If  $X$  is a continuous random variable with density  $f$ , and  $f$  is continuous at  $x$ , then  $f(x)$  is the limit of the probability that  $X$  falls in an interval starting at  $x$ , divided by the width of that interval, as that width shrinks to 0:*

$$f(x) = \lim_{\Delta \downarrow 0} \frac{\Pr(x \leq X < x + \Delta)}{\Delta}$$

### **i** Proof

*Proof.* For  $\Delta > 0$ :

$$\begin{aligned} \frac{\Pr(x \leq X < x + \Delta)}{\Delta} &= \frac{\Pr(x < X \leq x + \Delta)}{\Delta} \quad (\Pr(X = x) = \Pr(X = x + \Delta) = 0) \\ &= \frac{F(x + \Delta) - F(x)}{\Delta} \quad (\text{additivity, and the definition of the CDF}) \end{aligned}$$

By Theorem 0.6,  $F$  is differentiable at  $x$  with  $F'(x) = f(x)$ , so this difference quotient converges to  $f(x)$  as  $\Delta \downarrow 0$ .  $\square$

*Remark.* For small  $\Delta > 0$ ,  $f(x) \cdot \Delta \approx \Pr(x \leq X < x + \Delta)$ . Although a density is not unique (Theorem 0.1), this limit pins down its value at every point where it is continuous. See also Rothman et al. (2021) (Chapter 22, p. 535).

**Example 0.16** (The uniform density as a limit). For  $X \sim \text{Uniform}(0, 1)$  and  $x \in [0, 1]$ ,  $\Pr(x \leq X < x + \Delta) = \Delta$  once  $\Delta \leq 1 - x$ , so:

$$f(x) = \lim_{\Delta \downarrow 0} \frac{\Delta}{\Delta} = 1$$

For  $x < 0$  or  $x > 1$ , the interval eventually misses  $[0, 1]$ , so the limit is 0. Both agree with the density of Example 0.8.

**Proposition 0.1** (The density limit is the right derivative of the CDF). *If  $X$  is a continuous random variable with CDF  $F$ , then for every  $x$ :*

$$\lim_{\Delta \downarrow 0} \frac{\Pr(x \leq X < x + \Delta)}{\Delta} = \lim_{\Delta \downarrow 0} \frac{F(x + \Delta) - F(x)}{\Delta}$$

whenever either limit exists; the right-hand side is the right derivative of  $F$  at  $x$ .

### **i** Proof

*Proof.* The first two lines of the proof of Theorem 0.8 use only that  $X$  is continuous, so for every  $\Delta > 0$  the two quotients are equal. Equal functions of  $\Delta$  have the same limit as  $\Delta \downarrow 0$ , or both have none.  $\square$

**Example 0.17** (The density limit at a jump). At a point where  $f$  is not continuous, the limit can differ from the chosen value  $f(x)$ . For  $X \sim \text{Uniform}(0, 1)$  with the density  $f$  of Example 0.8, which sets  $f(1) = 1$  and  $f(x) = 0$  for  $x > 1$ , and for  $\Delta > 0$ :

$$\begin{aligned} \Pr(1 \leq X < 1 + \Delta) &= \Pr(1 \leq X \leq 1 + \Delta) \quad (\Pr(X = 1 + \Delta) = 0) \\ &= \int_1^{1+\Delta} f(x) dx \quad (\text{definition of the density}) \\ &= 0 \quad (f = 0 \text{ on } (1, 1 + \Delta]; \text{ one point does not change an integral}) \end{aligned}$$

So the limit of  $\Pr(1 \leq X < 1 + \Delta)/\Delta$  at  $x = 1$  is 0, although  $f(1) = 1$ . By Proposition 0.1, 0 is also the right derivative of  $F$  at 1, matching the slope of 0 on the right in Example 0.14.

**Theorem 0.9** (Density functions integrate to 1). *If  $X$  is a continuous random variable with density  $f$ :*

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

### **i** Proof

*Proof.*

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \lim_{b \rightarrow \infty} \int_{-\infty}^b f(x) dx \quad (\text{definition of an improper integral}) \\ &= \lim_{b \rightarrow \infty} F(b) \quad (\text{the CDF is the integral of the density}) \\ &= 1 \quad (\text{limit of a CDF}) \end{aligned}$$

The last step is Theorem 0.4.  $\square$

**Example 0.18** (The uniform density integrates to 1). For  $X$  uniform on  $[0, 1]$  (Example 0.8),  $\int_{-\infty}^{\infty} f(x) dx = \int_0^1 1 dx = 1$ .

**Definition 0.14** (Jointly distributed random variables). Random variables  $X_1, \dots, X_n$  are **jointly distributed** when they are defined on the same **sample space**  $\Omega$ , so that an event about several of them at once, such as  $\{X_1 \leq x_1\} \cap \{X_2 \leq x_2\}$ , has a probability.

*Remark.* We write  $\Pr(X \in A, Y \in B)$  for  $\Pr(\{X \in A\} \cap \{Y \in B\})$ , and similarly for more variables.

**Example 0.19** (The first flip and the total). In Example 0.1, let  $X_1$  indicate heads on the first flip (as in Example 0.7), and let  $X$  be the total number of heads. Both are functions on the same sample space,  $\{HH, HT, TH, TT\}$ , so they are jointly distributed, and, for example:

$$\Pr(X_1 = 1, X = 1) = \Pr(\{HH, HT\} \cap \{HT, TH\}) = \Pr(\{HT\}) = \frac{1}{4}$$

**Definition 0.15** (Joint distribution). The **joint distribution** of **jointly distributed** random variables  $X_1, \dots, X_n$  is the function that assigns to each set  $A \subseteq \mathbb{R}^n$  for which  $\{(X_1, \dots, X_n) \in A\}$  is an **event** the probability  $\Pr((X_1, \dots, X_n) \in A)$ .

**Example 0.20** (Joint distribution of the first flip and the total). In Example 0.19, the joint distribution of  $X_1$  and  $X$  assigns to the set  $A = \{(x_1, x) : x_1 = 1, x \geq 1\}$  the probability

$$\Pr((X_1, X) \in A) = \Pr(\{HH, HT\}) = \frac{1}{2}$$

since the first flip is heads exactly on  $HH$  and  $HT$ , and each of those outcomes has at least one head.

**Theorem 0.10** (A joint distribution is determined by its rectangle probabilities). If **jointly distributed** random variables  $X_1, \dots, X_n$ , and **jointly distributed** random variables  $Y_1, \dots, Y_n$ , satisfy

$$\Pr(X_1 \in A_1, \dots, X_n \in A_n) = \Pr(Y_1 \in A_1, \dots, Y_n \in A_n)$$

for all sets  $A_1, \dots, A_n$  of real numbers for which these are events, then  $(X_1, \dots, X_n)$  and  $(Y_1, \dots, Y_n)$  have the same **joint distribution**.

*Remark.* The proof needs measure theory beyond these notes: the sets  $A_1 \times \cdots \times A_n$  form a  $\pi$ -system, and two probability measures that agree on a  $\pi$ -system agree on the  $\sigma$ -algebra it generates (Billingsley 1995).

**Definition 0.16** (Joint probability mass function). For jointly distributed discrete random variables  $X$  and  $Y$ , the **joint probability mass function** (joint PMF) of  $X$  and  $Y$  is the probability that  $X$  takes the value  $x$  and  $Y$  takes the value  $y$ :

$$P(X = x, Y = y) \stackrel{\text{def}}{=} \Pr(\{X = x\} \cap \{Y = y\})$$

**Example 0.21** (Joint PMF of the first flip and the total). Continuing Example 0.19, each outcome of the two flips fixes both  $X_1$  and  $X$ , so the joint PMF puts probability  $1/4$  on each outcome's pair of values:

|           | $X = 0$        | $X = 1$        | $X = 2$        |
|-----------|----------------|----------------|----------------|
| $X_1 = 0$ | $1/4$ ( $TT$ ) | $1/4$ ( $TH$ ) | $0$            |
| $X_1 = 1$ | $0$            | $1/4$ ( $HT$ ) | $1/4$ ( $HH$ ) |

**Definition 0.17** (Marginal distribution). When a random variable  $X$  is one of several jointly distributed random variables, the distribution of  $X$  by itself (its CDF, and its PMF or density) is called the **marginal distribution** of  $X$ .

*Remark.* The word “marginal” only says that the other variables are being set aside; the marginal distribution of  $X$  is the same object as the distribution of  $X$ .

**Example 0.22** (Marginal distribution of the total). In Example 0.21, the marginal distribution of the total  $X$  is the PMF of Example 0.6:  $P(X = 0) = 1/4$ ,  $P(X = 1) = 1/2$ ,  $P(X = 2) = 1/4$ .

**Theorem 0.11** (Marginal PMF from a joint PMF). *If  $X$  and  $Y$  are jointly distributed discrete random variables, then for every  $x$ :*

$$P(X = x) = \sum_{y \in \mathcal{R}(Y)} P(X = x, Y = y)$$

**i** Proof

*Proof.* The event  $\{X = x\}$  is the disjoint union of the events  $\{X = x\} \cap \{Y = y\}$  over the countably many values  $y \in \mathcal{R}(Y)$ , so:

$$\begin{aligned} P(X = x) &= \Pr\left(\bigcup_{y \in \mathcal{R}(Y)} (\{X = x\} \cap \{Y = y\})\right) && \text{(the events partition } \{X = x\}) \\ &= \sum_{y \in \mathcal{R}(Y)} \Pr(\{X = x\} \cap \{Y = y\}) && \text{(countable additivity)} \\ &= \sum_{y \in \mathcal{R}(Y)} P(X = x, Y = y) && \text{(definition of the joint PMF)} \end{aligned}$$

□

**Example 0.23** (Summing a row of the joint PMF). In Example 0.21, summing the  $X_1 = 1$  row:

$$\begin{aligned} P(X_1 = 1) &= P(X_1 = 1, X = 0) + P(X_1 = 1, X = 1) + P(X_1 = 1, X = 2) && \text{(marginal PMF from the joint PMF)} \\ &= 0 + \frac{1}{4} + \frac{1}{4} && \text{(read the table)} \\ &= \frac{1}{2} && \text{(add)} \end{aligned}$$

which matches  $X_1 \sim \text{Ber}(1/2)$  from Example 0.7.

**Definition 0.18** (Joint probability density function). For [jointly distributed](#) continuous random variables  $X$  and  $Y$ , a **joint probability density function** (joint density) of  $X$  and  $Y$ , denoted  $f_{X,Y}(x, y)$  or  $p(X = x, Y = y)$ , is a function  $f_{X,Y}$  on  $\mathbb{R}^2$  that satisfies:

- $f_{X,Y}(x, y) \geq 0$  for every  $(x, y)$ .
- The integral of  $f_{X,Y}$  over any region  $A \subseteq \mathbb{R}^2$  is the probability that the pair  $(X, Y)$  falls in  $A$ :

$$\Pr((X, Y) \in A) = \iint_A f_{X,Y}(x, y) dx dy$$

*Remark.* As with [events](#), a fully rigorous version restricts  $A$  to a designated collection of regions; every region that arises in these notes is in it.

**Example 0.24** (A joint density on a triangle). Let  $f_{X,Y}(x, y) = 2$  for  $0 \leq x \leq y \leq 1$ , and 0 otherwise. The triangle  $\{(x, y) : 0 \leq x \leq y \leq 1\}$  has area  $1/2$ , so  $f_{X,Y}$  gives the

whole plane probability  $2 \cdot \frac{1}{2} = 1$ , and it is the joint density of a pair  $(X, Y)$  with  $X \leq Y$  always. For example, the probability that both are at most  $1/2$  is 2 times the area of the smaller triangle  $\{0 \leq x \leq y \leq 1/2\}$ :

$$\Pr(X \leq \tfrac{1}{2}, Y \leq \tfrac{1}{2}) = 2 \cdot \tfrac{1}{8} = \tfrac{1}{4}$$

**Theorem 0.12** (Not every pair of continuous random variables has a joint density). *If  $X$  is a continuous random variable, then the pair  $(X, X)$  has no joint density.*

### **i** Proof

*Proof.* Suppose  $f$  were a joint density of  $(X, X)$ , and let  $L = \{(x, y) : y = x\}$  be the diagonal line. Every outcome  $\omega$  has  $X(\omega) = X(\omega)$ , so  $\{(X, X) \in L\} = \Omega$ . For each  $x$ , the function  $y \mapsto 1_{y=x}f(x, y)$  is 0 except at the single point  $y = x$ , so its integral over  $y$  is 0. Then:

$$\begin{aligned} 1 &= \Pr((X, X) \in L) && (\{(X, X) \in L\} = \Omega, \text{ and } \Pr(\Omega) = 1) \\ &= \iint_L f(x, y) dx dy && (\text{definition of a joint density}) \\ &= \int_{-\infty}^{\infty} \left( \int_{-\infty}^{\infty} 1_{y=x} f(x, y) dy \right) dx && (\text{iterate the integral; Tonelli's theorem}) \\ &= \int_{-\infty}^{\infty} 0 dx && (\text{the inner integrand is 0 except at } y = x) \\ &= 0 && (\text{integrate}) \end{aligned}$$

which is a contradiction. Tonelli's theorem allows the iterated integral because  $f \geq 0$  ([Fubini–Tonelli theorem](#); Billingsley (1995), Theorem 18.3).  $\square$

**Theorem 0.13** (Marginal density from a joint density). *If  $X$  and  $Y$  have joint density  $f_{X,Y}$ , then  $X$  has the density:*

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

### **i** Proof

*Proof.* For  $a \leq b$ , the event  $\{a \leq X \leq b\}$  is the event that  $(X, Y)$  falls in the strip  $[a, b] \times \mathbb{R}$ , so:

$$\begin{aligned}
\Pr(a \leq X \leq b) &= \Pr((X, Y) \in [a, b] \times \mathbb{R}) && \text{(same event)} \\
&= \iint_{[a, b] \times \mathbb{R}} f_{X, Y}(x, y) dx dy && \text{(definition of a joint density)} \\
&= \int_a^b \left( \int_{-\infty}^{\infty} f_{X, Y}(x, y) dy \right) dx && \text{(iterate the integral; Tonelli's theorem)} \\
&= \int_a^b f_X(x) dx && \text{(definition of } f_X)
\end{aligned}$$

Tonelli's theorem allows the iterated integral because  $f_{X, Y} \geq 0$  ([Fubini–Tonelli theorem](#); Billingsley (1995), Theorem 18.3). So  $f_X$  satisfies Definition 0.8.  $\square$

**Example 0.25** (Marginal densities on the triangle). For the joint density of Example 0.24 and  $x \in [0, 1]$ ,  $f_{X, Y}(x, y) = 2$  exactly when  $x \leq y \leq 1$ , so:

$$\begin{aligned}
f_X(x) &= \int_{-\infty}^{\infty} f_{X, Y}(x, y) dy && \text{(marginal density from a joint density)} \\
&= \int_x^1 2 dy && (f_{X, Y}(x, y) = 2 \text{ for } x \leq y \leq 1, \text{ else } 0) \\
&= 2(1 - x) && \text{(integrate)}
\end{aligned}$$

The same steps with the roles swapped give  $f_Y(y) = \int_0^y 2 dx = 2y$  for  $y \in [0, 1]$ .

**Definition 0.19** (Joint density-mass function). Let  $X$  be a discrete random variable and  $Y$  a continuous random variable, [jointly distributed](#). A **joint density-mass function** of  $X$  and  $Y$ , denoted  $p(X = x, Y = y)$ , is a function  $p(X = x, Y = y) \geq 0$  that is a probability mass in  $x$  and a probability density in  $y$ : for every  $x$  and every set  $B \subseteq \mathbb{R}$ ,

$$\Pr(X = x, Y \in B) = \int_B p(X = x, Y = y) dy$$

*Remark.* When  $X$  is continuous and  $Y$  is discrete, the roles swap:  $\Pr(X \in B, Y = y) = \int_B p(X = x, Y = y) dx$ .

**Corollary 0.2** (Marginal PMF from a joint density-mass function). *If  $X$  and  $Y$  have joint density-mass function  $p(X = x, Y = y)$ , then for every  $x$ :*

$$P(X = x) = \int_{-\infty}^{\infty} p(X = x, Y = y) dy$$

### Proof

*Proof.* Every outcome has  $Y(\omega) \in \mathbb{R}$ , so  $\{Y \in \mathbb{R}\} = \Omega$ , and:

$$\begin{aligned} P(X = x) &= \Pr(\{X = x\} \cap \Omega) && (\{X = x\} \subseteq \Omega) \\ &= \Pr(X = x, Y \in \mathbb{R}) && (\{Y \in \mathbb{R}\} = \Omega) \\ &= \int_{-\infty}^{\infty} p(X = x, Y = y) dy && \text{(definition of a joint density-mass function, with } B = \mathbb{R}) \end{aligned}$$

□

**Example 0.26** (A coin flip and a waiting time). Let  $p(X = 0, Y = y) = 1/2$  for  $y \in [0, 1]$ ,  $p(X = 1, Y = y) = 1/4$  for  $y \in [0, 2]$ , and 0 otherwise. Integrating over  $y$  gives  $P(X = 0) = 1/2$  and  $P(X = 1) = 1/2$ , which add to 1. For example, the probability that  $X = 1$  and  $Y \leq 1$  is:

$$\Pr(X = 1, Y \leq 1) = \int_0^1 \frac{1}{4} dy = \frac{1}{4}$$

### Video lecture

Hutchinson's [Probability Refresher](#) (27 min) covers probability mass functions and probability density functions (Hutchinson, n.d.). The login for the video site is posted [on Canvas](#).

## Survival, hazard, and cumulative hazard functions

**Definition 0.20** (Survival function). The **survival function** (or **survivor function**) of a random variable  $T$ , denoted  $S(t)$ , is the probability that  $T$  exceeds  $t$ :

$$S(t) \stackrel{\text{def}}{=} \Pr(T > t)$$

*Remark.* The name comes from time-to-event analysis: if  $T$  is the time at which a participant dies, then  $S(t)$  is the probability that the participant is still alive at time  $t$ .

The definition itself applies to any random variable.

**Theorem 0.14** (Survival function and CDF). *For any random variable  $T$  with CDF  $F(t)$ :*

$$S(t) = 1 - F(t)$$

*If  $T$  is continuous with density  $f(t)$ , then also:*

$$S(t) = \int_{u=t}^{\infty} f(u) du$$

### **i** Proof

*Proof.* The event  $\{T > t\}$  is the complement of the event  $\{T \leq t\}$ , so the complement rule gives:

$$\begin{aligned} S(t) &\stackrel{\text{def}}{=} \Pr(T > t) && \text{(definition of the survival function)} \\ &= 1 - \Pr(T \leq t) && \text{(complement rule)} \\ &= 1 - F(t) && \text{(definition of the CDF)} \end{aligned}$$

For continuous  $T$ , the density integrates to 1 over the real line (Theorem 0.9), and  $F$  is the integral of the density up to  $t$  (Theorem 0.6), so:

$$\begin{aligned} 1 - F(t) &= \int_{u=-\infty}^{\infty} f(u) du - \int_{u=-\infty}^t f(u) du && \text{(total integral is 1; } F \text{ is the integral of } f) \\ &= \int_{u=t}^{\infty} f(u) du && \text{(split the first integral at } t \text{ and cancel)} \end{aligned}$$

□

**Example 0.27** (Survival function of an exponential distribution). Let  $T$  be exponential with rate  $\lambda > 0$  (Definition 0.11), so  $T$  has density  $f(t) = \lambda e^{-\lambda t}$  for  $t \geq 0$ . For  $t \geq 0$ , by Theorem 0.14:

$$\begin{aligned}
S(t) &= \int_{u=t}^{\infty} \lambda e^{-\lambda u} du \quad (\text{integral form of the survival function}) \\
&= [-e^{-\lambda u}]_{u=t}^{\infty} \quad (\text{antiderivative of } \lambda e^{-\lambda u}) \\
&= 0 - (-e^{-\lambda t}) \quad (\text{evaluate at the bounds; } e^{-\lambda u} \rightarrow 0 \text{ as } u \rightarrow \infty) \\
&= e^{-\lambda t} \quad (\text{simplify})
\end{aligned}$$

For  $t < 0$ ,  $S(t) = \Pr(T > t) = 1$ , since  $T \geq 0$ . With  $\lambda = 0.5$ , for example,  $S(2) = e^{-1} \approx 0.368$ .

**Definition 0.21** (Hazard function). The **hazard function** (also called the **hazard rate** or **hazard rate function**) of a continuous random variable  $T$  at a value  $t$  with  $\Pr(T \geq t) > 0$ , typically denoted  $\lambda(t)$  or  $h(t)$ , is the limit of the **conditional probability** that  $T$  falls in an interval starting at  $t$ , given  $T \geq t$ , divided by the width of that interval, as that width shrinks to 0:

$$\lambda(t) \stackrel{\text{def}}{=} \lim_{\Delta \downarrow 0} \frac{\Pr(t \leq T < t + \Delta \mid T \geq t)}{\Delta}$$

*Remark.* Sources differ on the symbol:  $h(t)$  appears in Dobson and Barnett (2018), Vittinghoff et al. (2012), Klein and Moeschberger (2003), and Kleinbaum and Klein (2012), while  $\lambda(t)$  appears in Rothman et al. (2021) and Kalbfleisch and Prentice (2011). If  $T$  is the time at which an event occurs, then  $\lambda(t)$  is a rate, not a probability: for a small interval width  $\Delta > 0$ ,  $\lambda(t) \cdot \Delta$  is approximately the probability that the event occurs in  $[t, t + \Delta)$ , given that it has not occurred before  $t$ . Many sources write the hazard as  $\lambda(t) = p(T = t \mid T \geq t)$ , reading it as the density of  $T$  at  $t$ , conditional on the event  $T \geq t$ ; that notation abbreviates the limit in Definition 0.21. For a discrete  $T$ , the analogous quantity is the **discrete-time hazard**.

The name “hazard” carries a connotation that the event is undesirable — death, relapse, equipment failure, and so on. When the event in question is neutral or desirable (recovery, conception, graduation, response to treatment), the same quantity  $\lambda(t)$  is often called the **event incidence rate** instead. This terminology parallels the convention that conditional probabilities of undesirable events are called **risks**, while the same conditional probabilities for neutral or desirable events are simply called **probabilities**. The math is identical; only the name changes with the valence of the event.

**Example 0.28** (A hazard greater than 1). A hazard can exceed 1, just as a density can (Example 0.9). Let  $T$  be **exponential** with rate  $\lambda = 2$ , so  $S(t) = e^{-2t}$  for  $t \geq 0$  (Example 0.27). Since  $T$  is continuous,  $\Pr(T = s) = 0$  for every  $s$ , so  $\Pr(T \geq s) = \Pr(T > s) = S(s)$ . For  $t \geq 0$  and  $\Delta > 0$ ,  $\{T \geq t\}$  is the disjoint union of  $\{t \leq T < t + \Delta\}$  and

$\{T \geq t + \Delta\}$ , so:

$$\begin{aligned}
\Pr(t \leq T < t + \Delta \mid T \geq t) &= \frac{\Pr(\{t \leq T < t + \Delta\} \cap \{T \geq t\})}{\Pr(T \geq t)} && \text{(definition of conditional probability)} \\
&= \frac{\Pr(t \leq T < t + \Delta)}{\Pr(T \geq t)} && \text{(subset property)} \\
&= \frac{\Pr(T \geq t) - \Pr(T \geq t + \Delta)}{\Pr(T \geq t)} && \text{(additivity)} \\
&= \frac{S(t) - S(t + \Delta)}{S(t)} && (\Pr(T \geq s) = S(s)) \\
&= \frac{e^{-2t} - e^{-2(t+\Delta)}}{e^{-2t}} && \text{(exponential survival function)} \\
&= 1 - e^{-2\Delta} && \text{(divide by } e^{-2t})
\end{aligned}$$

The [subset property](#) applies because  $\{t \leq T < t + \Delta\} \subseteq \{T \geq t\}$ . Then:

$$\begin{aligned}
\lambda(t) &= \lim_{\Delta \downarrow 0} \frac{1 - e^{-2\Delta}}{\Delta} && \text{(definition of the hazard function)} \\
&= \frac{\partial}{\partial \Delta} (1 - e^{-2\Delta}) \Big|_{\Delta=0} && \text{(definition of the derivative; } 1 - e^0 = 0) \\
&= 2e^0 && \text{(chain rule)} \\
&= 2 && (e^0 = 1)
\end{aligned}$$

So  $\lambda(t) = 2 > 1$  at every  $t \geq 0$ : a hazard is a rate, not a probability.

**Definition 0.22** (Discrete-time hazard). The **discrete-time hazard** of a [discrete](#) random variable  $T$  at a value  $t$  with  $\Pr(T \geq t) > 0$  is the [conditional probability](#) that  $T$  equals  $t$ , given  $T \geq t$ :

$$\Pr(T = t \mid T \geq t)$$

**Example 0.29** (Rolling until the first six). Roll a fair die repeatedly, and let  $T$  be the number of the roll that first shows a six. Then  $T \geq t$  exactly when the first  $t - 1$  rolls are not sixes, which has probability  $(5/6)^{t-1}$ , and  $T = t$  when, in addition, roll  $t$  is a six. So, for each  $t = 1, 2, \dots$ :

$$\begin{aligned}
\Pr(T = t \mid T \geq t) &= \frac{\Pr(T = t, T \geq t)}{\Pr(T \geq t)} && \text{(definition of conditional probability)} \\
&= \frac{\Pr(T = t)}{\Pr(T \geq t)} && (T = t \text{ implies } T \geq t) \\
&= \frac{(5/6)^{t-1} \cdot (1/6)}{(5/6)^{t-1}} && \text{(rolls are independent)} \\
&= \frac{1}{6} && \text{(cancel)}
\end{aligned}$$

The discrete-time hazard is the same at every roll: having gone without a six so far does not change the chance of a six on the next roll.

**Corollary 0.3** (A discrete-time hazard lies in  $[0, 1]$ ). *The [discrete-time hazard](#) of a discrete random variable  $T$  at a value  $t$  with  $\Pr(T \geq t) > 0$  satisfies:*

$$0 \leq \Pr(T = t \mid T \geq t) \leq 1$$

#### **i** Proof

*Proof.*  $\{T \geq t\}$  is the disjoint union of  $\{T = t\}$  and  $\{T > t\}$ , so, by additivity,  $\Pr(T \geq t) = \Pr(T = t) + \Pr(T > t) \geq \Pr(T = t)$ . Since  $\{T = t\} \subseteq \{T \geq t\}$ :

$$\begin{aligned}
\Pr(T = t \mid T \geq t) &= \frac{\Pr(\{T = t\} \cap \{T \geq t\})}{\Pr(T \geq t)} && \text{(definition of conditional probability)} \\
&= \frac{\Pr(T = t)}{\Pr(T \geq t)} && \text{(subset property)}
\end{aligned}$$

The numerator is non-negative and at most the positive denominator, so the ratio lies in  $[0, 1]$ .  $\square$

*Remark.* Unlike the continuous-time [hazard function](#), which is a rate and can exceed 1 (Example [0.28](#)), the discrete-time hazard is a probability.

**Theorem 0.15** (Hazard equals density over survival). *If  $T$  is a continuous random variable with [density](#)  $f(t)$  and [survival function](#)  $S(t)$ , then for every  $t$  with  $S(t) > 0$  at which  $f$  is continuous:*

$$\lambda(t) = \frac{f(t)}{S(t)}$$

**i** Proof

*Proof.* The proof uses three facts. First, for  $\Delta > 0$  the event  $\{t \leq T < t + \Delta\}$  is a subset of the event  $\{T \geq t\}$ , so intersecting them leaves  $\{t \leq T < t + \Delta\}$  (the [subset property](#)). Second,  $\Pr(T = t) = 0$  for a continuous  $T$ , so  $\Pr(T \geq t) = \Pr(T > t) = S(t)$ , which is positive. Third, because  $f$  is continuous at  $t$ , Theorem 0.8 gives  $f(t)$  as the limit of  $\Pr(t \leq T < t + \Delta)/\Delta$ .

$$\begin{aligned}
 \lambda(t) &\stackrel{\text{def}}{=} \lim_{\Delta \downarrow 0} \frac{\Pr(t \leq T < t + \Delta \mid T \geq t)}{\Delta} && \text{(definition of the hazard function)} \\
 &= \lim_{\Delta \downarrow 0} \frac{1}{\Delta} \cdot \frac{\Pr(\{t \leq T < t + \Delta\} \cap \{T \geq t\})}{\Pr(T \geq t)} && \text{(definition of conditional probability)} \\
 &= \lim_{\Delta \downarrow 0} \frac{1}{\Delta} \cdot \frac{\Pr(t \leq T < t + \Delta)}{\Pr(T \geq t)} && \text{(subset property)} \\
 &= \frac{1}{\Pr(T \geq t)} \cdot \lim_{\Delta \downarrow 0} \frac{\Pr(t \leq T < t + \Delta)}{\Delta} && (\Pr(T \geq t) \text{ does not depend on } \Delta) \\
 &= \frac{f(t)}{\Pr(T \geq t)} && \text{(density as a limit; } f \text{ is continuous at } t) \\
 &= \frac{f(t)}{S(t)} && (\Pr(T = t) = 0 \text{ for continuous } T)
 \end{aligned}$$

□

**Example 0.30** (Hazard function of an exponential distribution). Continuing Example 0.27, for  $t > 0$ , where  $f$  is continuous:

$$\begin{aligned}
 \lambda(t) &= \frac{f(t)}{S(t)} && \text{(hazard equals density over survival)} \\
 &= \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} && \text{(substitute the exponential density and survival function)} \\
 &= \lambda && \text{(cancel } e^{-\lambda t})
 \end{aligned}$$

At  $t = 0$ , where  $f$  jumps from 0 to  $\lambda$ , Definition 0.21 gives the same value directly:  $\Pr(0 \leq T < \Delta \mid T \geq 0)/\Delta = (1 - e^{-\lambda\Delta})/\Delta \rightarrow \lambda$  as  $\Delta \downarrow 0$ . So the exponential distribution has a constant hazard for  $t \geq 0$ , equal to its rate parameter; for  $t < 0$ ,  $f(t) = 0$ , so  $\lambda(t) = 0$ .

**Definition 0.23** (Cumulative hazard function). The **cumulative hazard function** of a continuous random variable  $T$ , often denoted  $\Lambda(t)$  or  $H(t)$ , is the integral of its [hazard function](#) up to  $t$ :

$$\Lambda(t) \stackrel{\text{def}}{=} \int_{u=-\infty}^t \lambda(u) du$$

**Example 0.31** (Cumulative hazard function of an exponential distribution). Continuing Example 0.30, the hazard is  $\lambda(u) = \lambda$  for  $u \geq 0$  and 0 for  $u < 0$ , so for  $t \geq 0$ :

$$\begin{aligned} \Lambda(t) &= \int_{u=-\infty}^0 0 du + \int_{u=0}^t \lambda du && \text{(split the integral at 0)} \\ &= 0 + \lambda t && \text{(integrate each piece)} \\ &= \lambda t && \text{(simplify)} \end{aligned}$$

and  $\Lambda(t) = 0$  for  $t < 0$ .

**Corollary 0.4** (Cumulative hazard of a non-negative random variable). *If  $T$  is a continuous random variable with  $T \geq 0$ , such as a time to event, then  $\lambda(u) = 0$  for every  $u < 0$ , and for every  $t \geq 0$ :*

$$\Lambda(t) = \int_{u=0}^t \lambda(u) du$$

### **i** Proof

*Proof.* Let  $u < 0$ . Every outcome has  $T(\omega) \geq 0 > u$ , so  $\{T \geq u\} = \Omega$ , which has probability 1  $> 0$ , and  $\lambda(u)$  is defined. For  $0 < \Delta \leq -u$ , the event  $\{u \leq T < u + \Delta\}$  is empty, because  $u + \Delta \leq 0$ , so:

$$\begin{aligned} \lambda(u) &= \lim_{\Delta \downarrow 0} \frac{\Pr(u \leq T < u + \Delta \mid T \geq u)}{\Delta} && \text{(definition of the hazard function)} \\ &= \lim_{\Delta \downarrow 0} \frac{\Pr(\{u \leq T < u + \Delta\} \cap \{T \geq u\})}{\Delta \cdot \Pr(T \geq u)} && \text{(definition of conditional probability)} \\ &= \lim_{\Delta \downarrow 0} \frac{\Pr(\emptyset \cap \Omega)}{\Delta \cdot \Pr(\Omega)} && \text{(the two events found above)} \\ &= \lim_{\Delta \downarrow 0} \frac{0}{\Delta \cdot 1} && (\emptyset \cap \Omega = \emptyset; \Pr(\emptyset) = 0; \Pr(\Omega) = 1) \\ &= 0 && \text{(simplify)} \end{aligned}$$

Then, for  $t \geq 0$ :

$$\begin{aligned}
\Lambda(t) &= \int_{u=-\infty}^t \lambda(u) du && \text{(definition of the cumulative hazard)} \\
&= \int_{u=-\infty}^0 \lambda(u) du + \int_{u=0}^t \lambda(u) du && \text{(split the integral at 0)} \\
&= \int_{u=-\infty}^0 0 du + \int_{u=0}^t \lambda(u) du && (\lambda(u) = 0 \text{ for } u < 0) \\
&= \int_{u=0}^t \lambda(u) du && \text{(integrate)}
\end{aligned}$$

□

*Remark.* The form in Corollary 0.4, with lower limit 0, is the form most survival-analysis texts use.

**Corollary 0.5** (Survival function from the cumulative hazard). *If  $T$  is a continuous random variable whose density  $f$  is continuous at all but finitely many points, then for every  $t$  with  $S(t) > 0$ :*

$$\Lambda(t) = -\log\{S(t)\}$$

and therefore:

$$S(t) = \exp\{-\Lambda(t)\} \tag{1}$$

### **i** Proof

*Proof.* Since  $S(t) = 1 - F(t)$  (Theorem 0.14) and  $F$  has derivative  $f$  wherever  $f$  is continuous (Theorem 0.6),  $S$  has derivative  $-f$  at those points. At each such  $u$  with  $S(u) > 0$ :

$$\begin{aligned}
\frac{d}{du}(-\log\{S(u)\}) &= -\frac{1}{S(u)} \cdot \frac{d}{du} S(u) && \text{(chain rule)} \\
&= -\frac{1}{S(u)} \cdot (-f(u)) && (S' = -f) \\
&= \frac{f(u)}{S(u)} && \text{(simplify)} \\
&= \lambda(u) && \text{(hazard equals density over survival)}
\end{aligned}$$

The function  $-\log\{S(u)\}$  is continuous, because  $S$  is continuous for a continuous  $T$ , and it has derivative  $\lambda(u)$  at all but finitely many points, so it is an antiderivative of  $\lambda$  for the

fundamental theorem of calculus. As  $u \rightarrow -\infty$ ,  $S(u) \rightarrow 1$ , so  $-\log\{S(u)\} \rightarrow 0$ . Therefore:

$$\begin{aligned}\Lambda(t) &\stackrel{\text{def}}{=} \int_{u=-\infty}^t \lambda(u) du && \text{(definition of the cumulative hazard)} \\ &= [-\log\{S(u)\}]_{u=-\infty}^t && (-\log S \text{ is an antiderivative of } \lambda) \\ &= -\log\{S(t)\} - 0 && \text{(evaluate at the bounds)} \\ &= -\log\{S(t)\} && \text{(simplify)}\end{aligned}$$

Exponentiating both sides of  $-\Lambda(t) = \log\{S(t)\}$  gives Equation 1.  $\square$

**Example 0.32** (Recovering the exponential survival function from its cumulative hazard). Continuing Example 0.31, for  $t \geq 0$ :

$$\begin{aligned}S(t) &= \exp\{-\Lambda(t)\} && \text{(survival function from the cumulative hazard)} \\ &= \exp\{-\lambda t\} && \text{(substitute } \Lambda(t) = \lambda t)\end{aligned}$$

which matches the survival function computed directly in Example 0.27.

Table 1: Probability distribution functions of a continuous random variable  $T$

| Name                                   | Symbols            | Definition                                                                     |
|----------------------------------------|--------------------|--------------------------------------------------------------------------------|
| Probability density function (PDF)     | $f(t), p(t)$       | $f \geq 0$ with<br>$\int_a^b f(u) du = \Pr(a \leq T \leq b)$                   |
| Cumulative distribution function (CDF) | $F(t)$             | $\Pr(T \leq t)$                                                                |
| Survival function                      | $S(t), \bar{F}(t)$ | $\Pr(T > t)$                                                                   |
| Hazard function                        | $\lambda(t), h(t)$ | $\lim_{\Delta \downarrow 0} \Pr(t \leq T < t + \Delta \mid T \geq t) / \Delta$ |
| Cumulative hazard function             | $\Lambda(t), H(t)$ | $\int_{u=-\infty}^t \lambda(u) du$                                             |
| Log-hazard function                    | $\eta(t)$          | $\log\{\lambda(t)\}$                                                           |

*Remark.* For a continuous random variable  $T \geq 0$  whose density is continuous at all but finitely many points, the following results connect the functions in Table 1:

- Theorem 0.14
- Theorem 0.15
- Corollary 0.4
- Corollary 0.5

Each arrow in the following diagram converts one function into the next:

$$\begin{aligned}
f(t) &\xleftarrow[\frac{S(t)\lambda(t)}{-S'(t)}]{} S(t) \xleftarrow[\exp\{-\Lambda(t)\}]{} \Lambda(t) \xleftarrow[\int_{u=0}^t \lambda(u) du]{} \lambda(t) \xleftarrow[\exp\{\eta(t)\}]{} \eta(t) \\
f(t) &\xrightarrow[\int_{u=t}^{\infty} f(u) du]{} S(t) \xrightarrow[-\log\{S(t)\}]{} \Lambda(t) \xrightarrow[\Lambda'(t)]{} \lambda(t) \xrightarrow[\log\{\lambda(t)\}]{} \eta(t)
\end{aligned}$$

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